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J. SIAM APPL. MATH.
Vol. 14, No. 3, May, 1966
Printed in U.S.A.

ON DIRECTED TREES AND DIRECTED k-TREES OF A DIGRAPH AND THEIR GENERATION*

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1. Introduction. The efficiency with which a complex electrical network that is passive and contains no mutual inductances may be analyzed by a digital computer depends largely upon the efficiency with which the trees of the corresponding graph are generated [1]–[6]. When the network is active it is known that the determinants and their cofactors of the node-admittance matrix and/or the loop-impedance matrix of an active electrical network can be evaluated by finding the set of all directed trees and the set of all directed 2-trees of the corresponding directed linear graph (or simply digraph) [7]. The efficient generation of all directed trees and 2-trees of a digraph is therefore important. Since an undirected graph may be transformed into a directed graph by representing each undirected edge by a pair of directed edges with opposite directions, the process of generating all trees of an undirected graph becomes a special case of the more general technique of generating all directed trees of a digraph.

This paper presents a method for generating directed trees, directed 2-trees and, in general, directed k-trees of a given digraph. The method may be considered as an extension of the idea proposed by Wang [8], Ting [9], and Wozniacki [10], and is readily adaptable for digital computation. In the last part of the paper, specific applications of the technique to various simple network structures are given.

2. Definitions and preliminary considerations. A *directed linear graph* $G(V, E)$ (or simply *digraph* G) consists of a set V of elements called *nodes* together with a set E of ordered pairs of the form (i, j), i and $j \in V$, called the *directed edges* (or simply *edges*) of the digraph; the node i is called the *initial node* and node j the *terminal node*. Together they are called the *endpoints* of the edge (i, j).

For any $i \in V$ the symbols $\rho(i)$ and $\rho^*(i)$ denote the cardinals of the sets of edges of G having i as initial and terminal nodes, respectively. They are called the *outgoing* and *incoming degrees* of G at i. A digraph is *regular o degree* k if $\rho(i) = \rho^*(i) = k$ for each $i \in V$. A digraph is *symmetric* if it has the property that if $(i, j) \in E$, then $(j, i) \in E$.

A digraph S is a *subgraph* of G if the nodes and edges of S are nodes and

* Received by the editors June 1, 1965, and in final revised form October 27, 1965.

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edges respectively of G and if each edge of S has the same initial node and the same terminal node in S as in G. If the node set and the edge set of S are both empty, then S is called the *empty graph* $\varnothing$ of G. A node not incident to any edge is called an *isolated node*. A digraph consisting only of isolated nodes is called the *null graph*. The empty graph is considered as a subgraph of every digraph. A *directed circuit* of G is a connected regular subgraph of degree 1. Two subgraphs are *disjoint* when they have no nodes in common. If S is a subgraph of G and each node of G is a node of S, then S is a *spanning subgraph* of G. A *directed path* P_{ij} is a sequence of the form $P_{ij} = (i, k_1)(k_1, k_2)(k_2, k_3)\cdots(k_m, j) \in E$ where i, j, and k_t, $t = 1, \cdots, m$, are nodes in V. It is not required that all the nodes of P_{ij} shall be distinct. If they are, P_{ij} is said to be a *directed simple path*. Again node i is called the *initial node* of P_{ij} and node j the *terminal node*. Both nodes i and j are referred to as *end nodes*. A directed circuit (or a directed path) is of *length* m if the number of edges contained in the directed circuit (directed path) is m.

If the order of two endpoints of an edge is immaterial, i.e., $(i, j) = (j, i)$, then the edge is called an *undirected edge*. An *undirected graph* is a graph in which every edge is undirected. To every directed graph $G(V, E)$, there is also a *corresponding undirected graph* $G_u(V_u, E_u)$ (or simply G_u) such that $V_u = V$ and $(i, j) = (j, i) \in E_u$ if either $(i, j) \in E$ or $(j, i) \in E$ or both. Usually, a digraph G can be represented equivalently by a geometric diagram: The nodes are indicated by small circles, while any two of them, i and j, are joined by an arrowheaded continuous curve, or even a straight line, from i to j if $(i, j) \in E$. Similarly, an undirected graph can also be represented equivalently by a geometric diagram such that there is a continuous curve from i to j if $(i, j) = (j, i) \in E_u$. More intuitively G_u of G can be obtained from G by omitting the arrowheads and combining the parallel curves. The geometric concept may be considered slightly general since each curve may be counted with a certain multiplicity. This will be excluded in this discussion. However, all the results obtained here can be extended easily, and the extension is almost trivial.

Sometimes, it is desirable to change an undirected graph G_u into a directed one G by the process of duplication: one replaces each undirected edge in G_u by two edges with the same endpoints; to the two edges in such a pair one assigns opposite orders (or directions). An *undirected path* P_{ij} of G_u is a sequence of undirected edges of G_u which has the same form as defined earlier for a directed path. G is said to be *connected* if, for any two nodes i and j, there exists an undirected path in G_u with these two nodes as end nodes. A *component* of G is a maximal connected subgraph of G. Let A be a subset of V; then the *sectional graph* $G[A]$ of G defined by A is the subgraph

whose node set is A and whose edges are all those edges in G which connect two nodes in A.

A *tree* t of G_u is a spanning subgraph which is connected and contains no undirected circuits. A spanning subgraph H of G_u is said to be a *k-tree* of G_u if H has k components and contains no undirected circuits. Some of the components may each consist of an isolated node. Very often, k-trees in which certain designated nodes are required to be in different components are used. For example, $t_{ab,cde,f}$ is the symbol for a 3-tree in which the nodes a, b are in one component, the nodes c, d, e in one component, and the node f in one component. A subgraph, denoted by t^r, of a digraph $G(V, E)$ is said to be a *directed tree* if and only if (1) the corresponding undirected graph of t^r forms a tree in the corresponding undirected graph G_u of G; and (2) $\rho(i) = 1$ for all $i \in V - r$ where r is called the *reference node* of G and $V - r$ is the node set with node r having been deleted from V, and $\rho(r) = 0$. A subgraph, denoted by $t^r_{r,i}$, of G is said to be a *directed 2-tree* if and only if (1) the corresponding undirected graph of $t^r_{r,i}$ is a 2-tree in the corresponding undirected graph G_u of G; (2) each of the two components of $t^r_{r,i}$ is a directed tree for some sectional graph of G; node r serves as a reference node for one of such directed trees; and node i for the other. The extension to *directed k-tree* is obvious. Again $t^r_{ra,jbc,k}$ is the symbol for a directed 3-tree in which the nodes r, a are in one component; the nodes j, b, c in one component; and the node k in one component. Nodes r, j, k are the reference nodes of the components.

For a given set A, $\mu(A)$ is used to denote the number of elements in A. For convenience, edges of a given digraph $G(V, E)$ will also be represented by letters e_i. Each subgraph Q of G may be expressed by the "product" or by juxtaposition of its elements provided that Q contains no isolated nodes; i.e., for example, a directed tree may be represented by $e_{i_1}e_{i_2}\cdots e_{i_{\mu(V)-1}}$. Two set-theoretic binary operations, "union" and "intersection" represented by $\cup$ and $\cap$, respectively, will be used. $g_1 \cup g_2$ is a subgraph of G consisting of all elements which are either in g_1 or in g_2 or in both, where g_1 and g_2 are two subgraphs of G. $g_1 \cap g_2$ is a subgraph consisting of all elements which are in both g_1 and g_2. Again if $g_1 \cup g_2$ contains no isolated nodes, then it may be represented by the product of its elements. Similarly for $g_1 \cap g_2$; i.e., for example, if $g_1 = e_1e_2e_3e_4$ and $g_2 = e_1e_4e_5e_6$, then $g_1 \cup g_2 = e_1e_2e_3e_4e_5e_6$ and $g_1 \cap g_2 = e_1e_4$.

3. Proposed algebra. Let D be the set of all subgraphs of a given digraph $G(V, E)$. Consider the set H of all subsets of D for which an *equivalence relation* "$=$" is defined. Assume in H there are two binary operations. One of these binary operations is called the "ring-sum" and is denoted by the symbol $\oplus$. The second of these operations is called "star" and is denoted

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by the symbol $*$. These two operations are defined to satisfy the following rules: for h_1, $h_2 \in H$,

$$h_1 \oplus h_2 = \{x \mid x \text{ is in } h_1, \text{ or in } h_2, \text{ but not in both}\},$$

$$h_1 * h_2 = \{x \cup y \mid x \in h_1, y \in h_2, \text{ and } x \cap y = \varnothing\}.$$

It can easily be shown that the operations defined here are commutative and associative. Furthermore, the set H forms an abelian group with respect to the ring-sum operation, and a commutative semigroup with unity with respect to the star operation. The zero element in H is the empty set of H and is denoted by $h_0 = \{\}$, while the identity element is $\{\varnothing\}$, where $\varnothing$ is the empty graph of G. For every $h \in H$, $h_0 * h$ is defined to be h_0; and the inverse of h relative to the ring-sum operation is itself, i.e., $h \oplus h = h_0$.

4. Generation of directed trees.

(a) *For symmetrical digraphs.* Assume that the given digraph $G(V, E)$ is symmetric, and all the nodes of V are labelled by one of the integers from 1 to $\mu(V)$. Let C_1 be the Cartesian product of the subsets S_k, i.e.,

$$C_1 = \prod_{\substack{k=1 \\ k \neq r}}^{\mu(V)} S_k,$$

where $S_k = \{(k, t) | (k, t) \in E \text{ and } t = 1, 2, \cdots, k-1, k+1, \cdots, \mu(V)\}$ for $k = 1, 2, \cdots, \mu(V)$, and $k \neq r$, where r is an arbitrary node of V. Thus C_1 is the set of all subgraphs of G with outgoing degree 1 except at r where it is 0. It follows that the set of elements of C_1 can in general be grouped into three disjoint subsets M_t, $t = 1, 2, 3$, such that

$M_1 = \{c_i | c_i \in C_1, \text{ and } c_i \text{ is a directed tree of } G\}$,

$M_2 = \{c_i | c_i \in C_1, \text{ and there exists at least one directed circuit of length 2 in } c_i\}$,

$M_3 = \{c_i | c_i \in C_1, \text{ and } c_i \text{ is not a directed tree of } G \text{ and contains no directed circuits of length 2}\}$.

Since there exists a one-to-one correspondence between directed trees of a symmetrical digraph G and trees of the corresponding undirected graph G_u of G, [7], it is simpler to consider G_u rather than G, i.e., one considers $(i, j) = (j, i)$ for all $(i, j) \in E$. Consequently:

LEMMA 1. *For each* $c \in M_2$, $\{y_1\} * \cdots * \{y_{\mu(V)-1}\} = h_0$, *where* $c = y_1 y_2 \cdots y_{\mu(V)-1}$.

LEMMA 2. $\{c_1\} \oplus \{c_2\} \oplus \cdots \oplus \{c_{\mu(M_3)}\} = h_0$, *where* $c_i \in M_3$ *for* $i = 1, 2, \cdots, \mu(M_3)$.

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Proof. Suppose c' is an element of C_1 which contains at least one directed circuit of length > 2; then $c' = R_1 \cup R_2$, where $R_1 = (i_1, i_2)(i_2, i_3)\cdots(i_k, i_1)$, $k > 2$, is a directed circuit of c' and R_2 is the subgraph which contains those elements of c' which have not been accounted in R_1. Since G is assumed to be symmetric, there always exists a unique $c'' \in C_1$ other than c' such that $c'' = R_3 \cup R_2$, where $R_3 = (i_1, i_k)\cdots(i_3, i_2)(i_2, i_1)$, $k > 2$, is a directed circuit of c''. Since each element of M_3 appears exactly twice in M_3, the lemma follows immediately.

THEOREM 1. *In a given symmetric digraph $G(V, E)$, let $X_1 = S_1 * S_2 * \cdots * S_{r-1} * S_{r+1} * \cdots * S_{\mu(V)}$, where $S_k = \{(k, t) | (k, t) \in E$ and $t = 1, 2, \cdots, k-1, k+1, \cdots, \mu(V)\}$ for $k = 1, 2, \cdots, r-1, r+1, \cdots, \mu(V)$, and $r \in V$ is the reference node of G. Also let $T^r = \{x_1\} \oplus \{x_2\} \oplus \cdots \oplus \{x_{\mu(X_1)}\}$, where $x_i \in X_1$ for $i = 1, 2, \cdots, \mu(X_1)$. Then the set of all distinct directed trees t^r of $G(V, E)$ is given by T^r provided (i, j) and (j, i) are considered identical in the operations.*

Proof. By Lemma 1 it is apparent that X_1 is the set union of M_1 and M_3. By Lemma 2 it follows that $M_1 = T^r$. This completes the proof of the theorem.

(b) *For arbitrary digraphs.* Now an arbitrary digraph $G(V, E)$ is considered, i.e., it is no longer required that G be symmetric. In order that the method discussed in (a) be used to generate all directed trees of an arbitrary digraph G, it is necessary that G be modified first. Let $G'(V', E')$ be the digraph obtained from $G(V, E)$ such that $V' = V$ and $E' = \{(i, j) |$ either (i, j) or $(j, i) \in E\}$. It is obvious that $G'(V', E')$ is obtained from G by adding some edges not in E. In order that one may distinguish the added edges from those in E, the following labeling process for edges in E' will be adopted: if both (i, j) and (j, i) are in E, let $(i, j) = e_k$, $(j, i) = e_k'$; if $(i, j) \in E$, and $(j, i) \in E'$ but not in E, let $(i, j) = e_t$ and $(j, i) = e_t''$, where k and t are some positive integers.

Suppose the edges of $G'(V', E')$ are labelled according to the rules just adopted. The set T'^r of all directed trees t'^r of $G'(V', E')$ is then obtained by Theorem 1 provided e_k and e_k', e_t and e_t'' are considered, respectively, as identical elements in the operations. It follows that if one eliminates all those elements of T'^r which contain at least one double primed letter e_t'', then the remaining elements of T'^r form the set of all directed trees T^r of $G(V, E)$.

(c) *Illustrative example.* Consider the digraph $G(V, E)$ of Fig. 1a. The corresponding labelled digraph $G'(V', E')$ is shown in Fig. 1b.

$$S_1 * S_2 * S_3 * S_4 = \{e_7'\} * \{e_1, e_3'', e_4, e_7\} * \{e_1', e_2, e_6\} * \{e_2', e_3, e_5\}$$

$$= \{e_7'\} * \{e_1, e_3'', e_4\} * \{e_1', e_2, e_6\} * \{e_2', e_3, e_5\};$$

$$T'^r = \{e_1 e_2 e_5 e_7', \quad e_1 e_2' e_6 e_7', \quad e_1 e_3 e_6 e_7', \quad e_1 e_5 e_6 e_7', \quad e_1' e_3'' e_5 e_7', \; e_2 e_3'' e_5 e_7',$$

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$$e_2'e_3''e_6e_7',\ e_3''e_5e_6e_7',\ e_1'e_2'e_4e_7',\ e_1'e_3e_4e_7',\ e_1'e_4e_5e_7',\ e_2e_3e_4e_7',$$
$$e_2e_4e_5e_7',\ e_2'e_4e_6e_7',\ e_3e_4e_6e_7',\ e_4e_5e_6e_7'\};$$

$$T^r = \{e_1e_2e_5e_7',\ e_1e_2'e_6e_7',\ e_1e_3e_6e_7',\ e_1e_5e_6e_7',\ e_1'e_2'e_4e_7',$$
$$e_1'e_3e_4e_7',\ e_1'e_4e_5e_7',\ e_2e_3e_4e_7',\ e_2e_4e_5e_7',\ e_2'e_4e_6e_7',$$
$$e_3e_4e_6e_7',\ e_4e_5e_6e_7'\}.$$

Since the edges of the form (r, j) will never appear in any directed tree, the labeling for these edges is actually unnecessary, and can be ignored in actual computation.

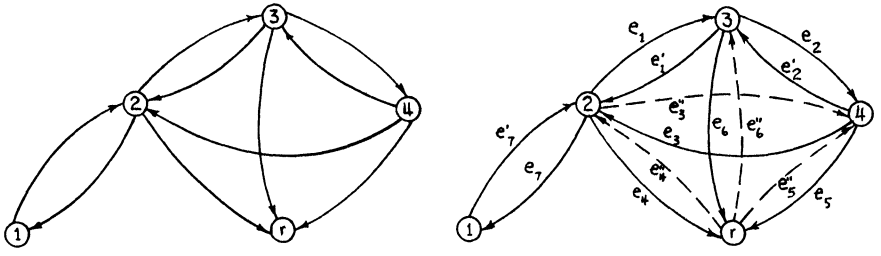
It should be noted that the expansion of $S_1*S_2*S_3*S_4$ is very simple. It amounts to forming all combinations of edges without duplication of subscripts. Since the star operation is associative, the number of combinations to be examined may be greatly reduced by eliminating some of the elements at an earlier stage; for example, if in S_1*S_2 there is an element with duplicated subscripts, then the element can be eliminated from S_1*S_2 before one forms the star operation with (S_3*S_4). Similarly, T'^r can be obtained from $S_1*S_2*S_3*S_4$ by eliminating all those elements in which the same subscripts appear elsewhere as a permutation. Again this step may be greatly simplified by eliminating those elements with same subscripts at an earlier stage; for example, if in S_1*S_2 there are two elements with the same subscripts then they can be eliminated from S_1*S_2 before one forms the star operation with S_3*S_4.

5. Generation of directed k-trees.

(a) *For symmetrical digraphs.* Let C_2 be the Cartesian product of the subsets S_j, i.e.,

$$C_2 = \prod_{\substack{j=1 \\ j\neq i_1,\cdots,i_{k-1},r}}^{\mu(V)} S_j,$$

where $S_j = \{(j, t) | (j, t) \in E$, and $t = 1, 2, \cdots, j-1, j+1, \cdots, \mu(V)\}$ for $j = 1, 2, \cdots, \mu(V)$ but $j \neq i_1, i_2, \cdots, i_{k-1}$, and r. It follows that each



(a) $G(V, E)$ (b) $G'(V', E')$

FIG. 1. *A given digraph $G(V, E)$ and its $G'(V', E')$*

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element c of C_2 is either a directed k-tree of the form $t_{r,i_1,\cdots,i_{k-1}}$ or a subgraph which contains at least one directed circuit of length $\geqq 2$. By eliminating the latter from C_2 it can easily be shown that the remaining elements of C_2 will give all possible directed k-trees of the form $t_{r,i_1,\cdots,i_{k-1}}$ of G. Thus, one may state:

THEOREM 2. *In a given symmetric digraph $G(V, E)$, let X_2 represent the star operation of the sets S_j, $j = 1, 2, \cdots, \mu(V)$ but $j \neq i_1, i_2, \cdots, i_{k-1}$ and r, where $S_j = \{(j, t)|(j, t) \in E$, and $t = 1, 2, \cdots, j-1, j+1, \cdots, \mu(V)\}$. Also let $T_k^r = \{x_1\} \oplus \{x_2\} \oplus \cdots \oplus \{x_{\mu(X_2)}\}$, where $x_i \in X_2$ for $i = 1, 2, \cdots, \mu(X_2)$. Then the set of all distinct directed k-trees $t_{r,i_1,\cdots,i_{k-1}}$, up to a null graph, is given by T_k^r provided (i, j) and (j, i) are considered identical in the operations.*

(b) *For arbitrary digraphs.* Suppose $G'(V', E')$ is obtained from $G(V, E)$ by the process outlined in §4(b) and is labelled accordingly. The set $T_k'^r$ of all directed k-trees of $G'(V', E')$ is then obtained by Theorem 2 provided e_k and e_k', e_t and e_t'', are considered, respectively, as identical elements in the operations. If those elements of $T_k'^r$ which contain at least one double primed letter e_t'' are eliminated, the remaining elements of $T_k'^r$, up to a null graph, are the set of all directed k-trees of the form $t_{r,i_1,\cdots,i_{k-1}}$ of $G(V, E)$.

(c) *Illustrative example.* Consider the digraph of Fig. 1. The set T_2^r of directed 2-trees of the form $t_{r,3}^r$ is obtained as follows:

$$S_1 * S_2 * S_4 = \{e_7'\} * \{e_1, e_3'', e_4, e_7\} * \{e_2', e_3, e_5\}$$

$$= \{e_1e_2'e_7', e_1e_3e_7', e_1e_5e_7', e_2'e_3''e_7', e_3''e_5e_7', e_2'e_4e_7', e_3e_4e_7', e_4e_5e_7'\};$$

$$T_2^r = \{e_1e_2'e_7', e_1e_3e_7', e_1e_5e_7', e_2'e_4e_7', e_3e_4e_7', e_4e_5e_7'\}.$$

6. Applications.

(a) *Topological analysis for active electrical networks.* Many methods of developing topological formulae for general electrical networks are known [11]–[13]. Some of these features might be undesirable; for example, they may have too many rules [11], or might require finding the signs for each term [12]–[13]. Recently, a method of evaluating general network determinants and their cofactors by means of directed trees and directed k-trees was proposed by Chen [7] which shows considerable improvement over the old ones. For completeness, the main results of Chen are collected here as Theorem 3 and Theorem 4.

Let $A_{n\times n}$ be the network matrix (either the node-admittance or the loop-impedance matrix) of a given network, where A is of the form

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$$A = \begin{bmatrix} a_{11} & -a_{12} & -a_{13} & \cdots & -a_{1n} \\ -a_{21} & a_{22} & -a_{23} & \cdots & -a_{2n} \\ & & \ddots & & \\ -a_{n1} & -a_{n2} & -a_{n3} & \cdots & a_{nn} \end{bmatrix}$$

which, in turn, can be written in a slightly different form

$$A = \begin{bmatrix} \sum_{x=1}^{n} b_{1x} & -b_{12} & -b_{13} & \cdots & -b_{1n} \\ -b_{21} & \sum_{x=1}^{n} b_{2x} & -b_{23} & \cdots & -b_{2n} \\ & & \ddots & & \\ -b_{n1} & -b_{n2} & -b_{n3} & \cdots & \sum_{x=1}^{n} b_{nx} \end{bmatrix},$$

where

$$b_{ij} = \begin{cases} a_{ij} & \text{for } i \neq j, \quad i, j = 1, \cdots, n, \\ a_{ij} - \sum_{\substack{x=1 \\ x \neq i}}^{n} a_{ix} & \text{for } i = j, \quad i, j = 1, \cdots, n. \end{cases}$$

The process of associating a digraph $G(V, E)$ with the above matrix is as follows:

1. $\mu(V) = n + 1$. Each node is labelled by one of the integers from 1 to n plus a node r called the reference node such that the node labelled k is associated with the kth row of A.
2. If $b_{ij} \neq 0$, $i \neq j$, there is an edge $(i, j) \in E$ with weight b_{ij}.
3. If $b_{ij} \neq 0$, $i = j$, there is an edge $(i, r) \in E$ with weight b_{ij}.

It should be noted that b_{ii} is the sum of the elements in the ith row of A and usually has very simple form. In many cases the associated digraph may be drawn directly from the given electrical network diagram without first writing out the corresponding network matrix.

From an abstract point of view, an electrical network is a weighted digraph. For convenience, the associated digraph will be described as a 3-tuple, denoted by $G(V, E, f)$, where f is a mapping function with domain in E and range in the complex field and such that $f((i, j))$ is the weight associated with the edge (i, j) of $G(V, E)$. Also it is convenient to extend f from a single edge $(i, j) \in E$ to any subgraph $R = \{x_i\}$ of $G(V, E, f)$ such that $f(R) = \pi f(x_i)$, where the mapping of an isolated node is defined to be 1 and the product is taken over all elements of R. If $Q = \{R_i\}$ is a set of subgraphs of $G(V, E, f)$ then $f(Q) = \{f(R_i)\}$; note that f is extended from

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a scalar-valued function over subgraphs to a vector-valued function over sets of subgraphs.

THEOREM 3. *With all the notation used above, the network determinant is obtained as follows*:

$$\det A = \sum_{t^r} f(t^r),$$

where t^r *is a directed tree in* $G(V, E, f)$, *and the summation is taken over all* $t^r \in G(V, E, f)$.

THEOREM 4. *With all the notation used above, the* i, jth *cofactor* Δ_{ij} *of* $\det A$ *is obtained as follows*:

$$\Delta_{ij} = \sum_{t^r_{ij,r}} f(t^r_{ij,r}),$$

where $t^r_{ij,r}$ *is a directed 2-tree in* $G(V, E, f)$, *and the summation is taken over all* $t^r_{ij,r} \in G(V, E, f)$.

Consider the electrical network shown in Fig. 2, which represents a general potential feedback amplifier. The associated digraph $G(V, E)$ and $G'(V', E')$ of $G(V, E)$ are shown in Fig. 1. The set of weights is given by

$$f(e_7) = f(e_7') = y_1 ; \quad f(e_1) = y_f ; \quad f(e_1') = y_f - g_m ; \quad f(e_2) = y_p + g_m ;$$
$$f(e_2') = y_p ; \quad f(e_3) = g_m ; \quad f(e_4) = y_g ; \quad f(e_6) = y_2 ; \quad f(e_5) = y_k .$$

Hence, the output impedance Z_0 looking from the points p and r is given by

$$Z_0 = \frac{\Delta_{33}}{\Delta} = \frac{(y_f + y_g)(y_p + g_m + y_k)}{(y_k + y_p + g_m)(y_f y_2 + y_f y_g + y_g y_2) + y_k(y_f y_p + y_p y_g + y_f g_m)}.$$

(b) *Topological analysis for nonrecurrent ladder networks.* Since the attempts of early pioneers in linear circuit analysis to formalize the approach to circuit problems, very little seems to have been subsequently achieved

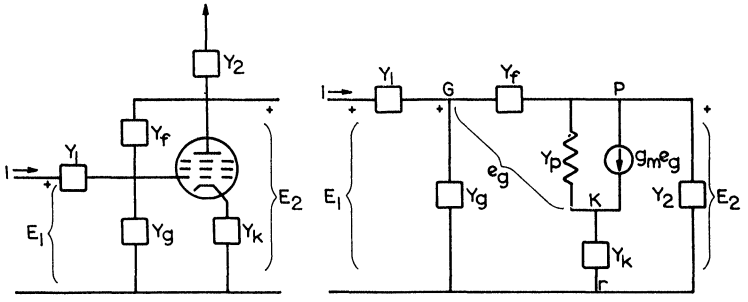


FIG. 2. *A general potential feedback amplifier and its equivalent circuit*

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towards developing rapid methods of network analysis. The method discussed above permits a rapid analysis of the nonrecurrent ladder networks which may contain active elements.

Since the network function can be expressed as the quotient of a determinant and its cofactor which in turn can be obtained from the sets of all directed trees and directed 2-trees of the corresponding digraph, the problem of network analysis may be greatly simplified.

7. Conclusions. A method, which is readily adaptable for digital computation for generating directed trees and directed k-trees of a given digraph, is presented. The method, in general, is easy to remember and simple to operate. When the digraph is symmetric, it reduces to the undirected case. Specific applications of the technique to the analysis of active electrical networks are also discussed.

The main advantage of this method lies in the fact that once a desired set of sets, which can be obtained easily by inspection, is derived from a given digraph all the remaining manipulations are algebraic. It is not necessary for one to go back to the given digraph to see whether a generated term forms a directed tree (k-tree) or not.

The ability of the method to reduce computational labor in evaluation of network determinants is of secondary importance. The analysis discussed above may provide a tool for the investigation of presently unsolved topological synthesis problems.

8. Acknowledgments. The author wishes to express his thanks to Dr. J. Miro for reading the manuscript and making many useful suggestions. He would also like to thank the National Science Foundation and the Ohio University Research Committee for the support of this research.

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